

Exercise I

Network estimation from time series data

PSYNETS

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Conceptual Questions

Question 1

Are the following statements true or false? Explain why.

1. If ergodicity holds, results from cross-sectional analysis are expected to equal results from within-person analysis.
2. It is generally recommended to always remove trends (such as linear trends) prior to analyzing your $N = 1$ time series.

Question 2

Suppose a therapist measures a patient about 75 times on a set of depression symptoms, including a question on “suicidal ideation”. Suppose that you estimate a graphical VAR model from this data and find that the node “suicidal ideation” is not connected to any of the other nodes in your network, neither in the temporal nor in the contemporaneous network. Think of three potential reasons why the node “suicidal ideation” may be disconnected in the resulting network.

$N = 1$ networks

Install packages

Make sure to install all required packages:

```
# dplyr
if (!requireNamespace("dplyr", quietly = TRUE)) {
  install.packages("dplyr")
}
library(dplyr)

# graphicalVAR
if (!requireNamespace("graphicalVAR", quietly = TRUE)) {
  install.packages("graphicalVAR")
}
library(graphicalVAR)

# psychonetrics
if (!requireNamespace("psychonetrics", quietly = TRUE)) {
  install.packages("psychonetrics")
}
library(psychonetrics)

# qgraph
if (!requireNamespace("qgraph", quietly = TRUE)) {
  install.packages("qgraph")
}
library(qgraph)

# patchwork
if (!requireNamespace("patchwork", quietly = TRUE)) {
  install.packages("patchwork")
}
```

```

}
library(patchwork)

# ggplot2
if (!requireNamespace("ggplot2", quietly = TRUE)) {
  install.packages("ggplot2")
}
library(ggplot2)

```

Explore the data

In this practical, you will analyze data collected by Fried, Papanikolaou, and Epskamp (2020) during the COVID-19 pandemic. For a detailed description of the data, analyses, and results, have a look at their [article](#) and [blogpost](#).

The dataset contains ESM time series data of 80 Dutch students, measured on items related to mental health measured four times a day over the course of two weeks (leading to 56 measurement occasions per individual). We will use this data to practice with estimating N=1 networks.¹ The items included in the study are the following:

Item	Abbreviation	Description
1	id	ID of the participant
2	Relax	I found it difficult to relax
3	Irritable	I felt (very) irritable
4	Worry	I was worried about different things
5	Nervous	I felt nervous, anxious or on edge
6	Future	I felt that I had nothing to look forward
7	Anhedonia	I couldn't seem to experience any positive feeling at all
8	Tired	I felt tired
9	Alone	I felt like I lack companionship, or that I am not close to people
10	Social_offline	I spent ...on meaningful, offline, social interaction
11	Social_online	I spent ...using social media to kill/pass the time
12	Outdoors	I spent ...outside (outdoors)
13	C19_occupied	I spent ...occupied with the coronavirus (e.g. watching news thinking about it talking to friends about it)
14	C19_worry	I spent ...thinking about my own health or that of my close friends and family members regarding the coronavirus
15	Home	I spent ...at home (including the home of parents/partner)
16	day	Day of the study on which the response was made
17	beep	Beep (measurement occasion) of the day on which the response was made
18	conc	Unique beep number on which the response was made

Note: Range of all items 1-5. Items 1 through 8: not at all, slightly, moderately, very, extremely. Items 9 through 14: 0 minutes, 1-15 minutes, 15-60 minutes, 1-2 hours, over 2 hours.

Download the file `clean_network.RData` from [OSF](#) and read into R:

```
load("clean_network.RData")
```

Next, run the following code to improve the node labels and remove some redundant items:

```

# Variables to investigate:
vars <- paste0("Q",1:18)

# Labels:
varLabs <- c("Relax","Irritable","Worry","Nervous","Future","Anhedonia",
             "Tired","Hungry","Alone","Angry","Social_offline","Social_online",
             "Music","Procrastinate","Outdoors","C19_occupied","C19_worry",
             "Home")

# Rename columns in data:

```

¹Note that in the original study, the authors used some advanced data processing. The results you will obtain here are based on simplified data processing and will therefore not exactly reproduce the findings of Fried, Papanikolaou, and Epskamp (2020).

```
names(Data2)[names(Data2) %in% vars] <- varLabs

# Remove items:
Data2 <- Data2 %>% select(-Hungry,-Angry,-Music,-Procrastinate)
varLabs <- varLabs[!varLabs %in% c("Hungry", "Angry", "Music", "Procrastinate")]
```

You now have two objects in your R environment: `Data2` containing the data, and `varLabs` containing the names of the variables.

To practice $N = 1$ time series analysis, we will load data of a single participant. For example, fill in your date of birth in the code below to select a random participant:

```
birth_date <- 07021996 # Enter date of birth here
set.seed(birth_date)
subject <- sample(Data2$id, 1)
my_data <- Data2[which(Data2$id == subject),]
```

Note: If your analysis leads to a lot of errors, this is possibly due to low variance in the chosen subject. You can also use a different seed such as the birthday of a fellow participant.

Next, choose 3 or 4 variables you are interested in (because the number of observations is quite low for each subject, we likely do not want to include many more than 4 variables).

```
# Example of how to select variables to include in the network:
vars <- c("C19_occupied", "C19_worry", "Alone")
```

Estimate GVAR using graphicalVAR

Question 3

Estimate a GVAR model on your sampled subject using the 3 or 4 variables you chose above using the *graphicalVAR* package (set $\gamma = 0$). Plot the estimated temporal (PDC) and contemporaneous (PCC) networks using the same layout, e.g., by creating an average layout with the `averageLayout()` function, and the same maximum. This is important for visual comparison, as edge width and saturation are scaled relative to the maximum edge weight.

Question 4

Repeat the above process for another subject. Use the same 3 or 4 variables as before. Plot the networks using the same layout and maximum as in Question 3. Compare the estimated networks. Can you conclude that there is heterogeneity (i.e., individual differences) between the two subjects based on their network structures? Why (not)?

Estimating GVAR with psychonetrics

Question 5

Estimate a *saturated* (i.e., fully connected) GVAR model on your sampled subject using the same 3 or 4 variables as before using the *psychonetrics* package. Plot the estimated temporal (PDC) and contemporaneous networks (`omega_zeta`) using the same layout and maximum.

Note: You can safely ignore warnings on non-positive definite matrices as long as the parameters look reasonable (e.g., no partial correlations near -1 and 1). The *psychonetrics* package models temporal effects using a matrix called *beta*, but also returns its standardized version called *PDC*. The PDC matrix is the standardized transpose of *beta* and can be plotted as a directed network.

Question 6

Inspect which of the estimated edges are significant. You can use the `CIplot()` function or inspect the parameters directly from the estimated model. Which edges are statistically significant?

Question 7

Now prune the saturated network model estimated with the *psychonetrics* package in Question 5 using an alpha rate of 0.05, with `recursive = FALSE`. Plot the resulting networks using the same layout and maximum as before.

Compare the pruned network to the one obtained in Question 3 using the *graphicalVAR* package. What are the differences and similarities?

Detrending

Linear trends over time (e.g., increasing or decreasing worry) can inflate spurious connections in network models. In this exercise, we test each variable for time trends and remove them if significant before re-estimating GVAR models.

We can use the time variable (in this case we use `conc` as a proxy for the measurement occasion) to check for trends in the data. To test if the variable “C19_worry” features a linear trend over time, we can run a linear regression using time as a predictor:

```
lm_c19_worry <- lm(C19_worry ~ conc, data = my_data)
summary(lm_c19_worry)
```

```
##
## Call:
## lm(formula = C19_worry ~ conc, data = my_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.03552 -0.30191 -0.01976  0.16423  1.32564
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.103235   0.152314   13.81 < 2e-16 ***
## conc        -0.022572   0.004542   -4.97 8.58e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5011 on 49 degrees of freedom
## (5 observations deleted due to missingness)
## Multiple R-squared:  0.3351, Adjusted R-squared:  0.3216
## F-statistic: 24.7 on 1 and 49 DF, p-value: 8.576e-06
```

This effect is significant. We can now “detrend” the variable worry as follows:

```
my_data$C19_worry[!is.na(my_data$C19_worry)] <- residuals(lm_c19_worry)
```

Question 8

Test for significant trends ($\alpha = 0.05$) for your selected variables and detrend these variables if significant trends were detected. Next, re-estimate a the network on the detrended data using either the `graphicalVAR` or `psychonetrics` package. Compare the networks you obtain after detrending to the ones estimated on the data before detrending. You want to use the same layout and maximum for a fair visual comparison. Which edges disappeared or appeared? What might this tell you about the role of time trends in your data?

Comparing $N = 1$ networks

When comparing the network structures of two individuals, visual inspection can be misleading. We might see differences that are due to chance or sample size, not true individual differences.² The Individual Network Invariance Test (INIT) is a statistical method that formally tests whether two networks are significantly different. It compares a model assuming network homogeneity (equality constraints) to a model allowing heterogeneity.

INIT is a developmental R package you can install from GitHub. Here’s how to install the package:

```
if (!require(remotes)) install.packages("remotes")
remotes::install_github("RiaHoekstra/INIT")
library(INIT)
```

The INIT package includes a sample data file containing simulated time series data with six variables (V1-V6) for two individuals (identified by `id`). Each individual’s data spans 100 time points ($t = 100$). Use the following line of code to load this dataset into R and store it in an object named `data`:

```
# Load data of two individuals:
data <- INIT::data
```

At a minimum, the INIT function requires you to specify the following arguments: `data`, which is the dataset, `idvar`, the variable identifying different individuals, and `vars`, the variables included in the analysis. The following line of code demonstrates how to use INIT with the specified arguments:

²If you want to delve deeper into this topic, you may find this article useful: [Hoekstra et al., \(2023\)](#).

```
# Compare networks with minimum required arguments:
INIT(data = data, idvar = colnames(data)[1], vars = colnames(data)[2:7])
```

```
##
## ----- INIT results -----
##
## The model without equality constraints is preferred.
##
## INIT summary
##   - AIC homogeneity model: 6847.434
##   - AIC heterogeneity model: 6771.743
##   - delta AIC: -75.69085
##
## input summary
##   - number of variables in the network: 6
##   - estimator used: ML
##   - estimated network type: saturated
##   - homogeneity test: overall
```

By default, INIT compares both the temporal and contemporaneous network structures of individuals estimated using maximum likelihood estimation and pruned at an alpha level of 0.05. However, these default settings can be adjusted if you, for example, only wish to compare the temporal network models between individuals. In addition, you can save the estimated network models for further inspection and visualization. To do this, set the `save_models` argument to `TRUE` when running the analysis.³ Store the results in an object as shown in the following example:

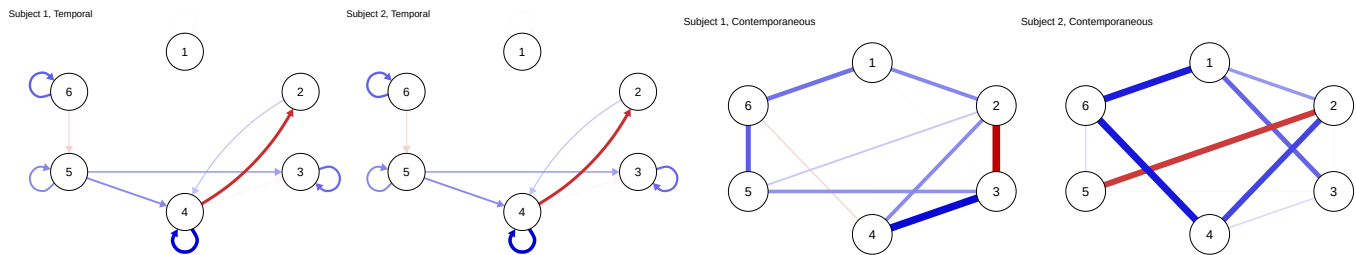
```
# Save the estimated network structures:
res <- INIT(data = data,
            idvar = colnames(data)[1],
            vars = colnames(data)[2:7],
            estimator = "FIML",
            network_type = "pruned",
            homogeneity_test = "temporal",
            save_models = TRUE)
```

```
res
```

```
##
## ----- INIT results -----
##
## The model with equality constraints is preferred.
##
## INIT summary
##   - BIC homogeneity model: 7150.666
##   - BIC heterogeneity model: 7155.727
##   - delta BIC: 5.060668
##
## input summary
##   - number of variables in the network: 6
##   - estimator used: FIML
##   - estimated network type: pruned
##   - homogeneity test: temporal
##
## Estimated networks are stored in x$network
```

After, we can plot the resulting networks using the `qgraph` packages:

³The INIT package relies on the `psychometrics` package, which we previously used to estimate GVAR networks for individuals. This means that the same arguments you employ in the `psychometrics` package to estimate a GVAR network for an individual can also be used with INIT when comparing the network structures of two individuals. For example, the `beepvar` and `dayvar` arguments can be used in a similar fashion in the INIT package.



Question 9

In Question 3 and 4 you estimated a GVAR network for two individuals containing 3 or 4 variables from the `Data2` object, and visually compared the resulting networks. Now, let's apply INIT to formally test the differences between these networks.

Since INIT requires a single data file as input, we first create a datafile containing the data for the two individuals you selected for Question 3 and 4 into one dataset. For example, here we merge the data of participant 68 and 32 in one dataset and select the relevant variables:

```
# Prepare your data:
my_data_init <- Data2[which(Data2$id == 68 | Data2$id == 32),]

# Define variables:
vars <- c("C19_occupied", "C19_worry", "Alone")
```

Now that we prepared our data set and selected our variables, apply INIT and inspect the results. Are your conclusions regarding individual differences the same as for Question 4? Why (not)?

Sample size determination

Simulation studies are a powerful tool within the field of network psychometrics, providing insight into sample size determination and best practices for achieving accurate results. In this part of the exercise we will practice with simulating time series data based on a GVAR model. In this [preprint](#) we explain the rationale behind simulation studies and present a step-by-step roadmap to initiate simulation studies. In addition, it provides an overview of open-source software packages available for simulating network models in the statistical programming environment R.

Using the `randomGVARmodel` function from the `graphicalVAR` package we can generate a random graphical VAR structure. This function simply requires the specification of the number of nodes for which we want to generate a graphical VAR network. See `?graphicalVAR` for the help file. However, given that the default probability for an edge in the contemporaneous network is low (i.e., 0.1), in addition to the number of nodes, we specify the `probKappaEdge` argument to create a more densely connected contemporaneous network:

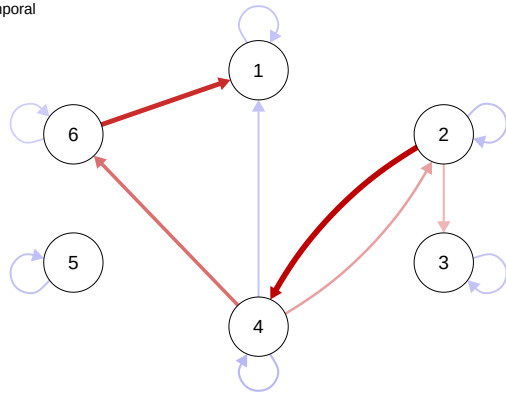
```
set.seed(5)
true_model <- randomGVARmodel(6, probKappaEdge = 0.6)
```

The `randomGVARmodel` function returns 4 model matrixes saved in the object `true_model`: The residual inverse variance-covariance matrix $Kappa$, the $beta$ matrix, the partial contemporaneous correlations (PCC) and the partial temporal correlations (PDC). To visually inspect what our model looks like we can plot the PDC matrix and the PCC matrix:

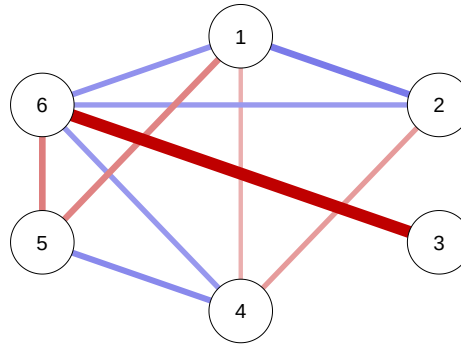
```
# Inspect generated model:
qgraph(true_model$PDC,
  title = "temporal",
  theme = "colorblind",
  vsize = 11,
  asize = 5,
  mar = rep(6,4),
  layout = "circle")

qgraph(true_model$PCC,
  title = "contemporaneous",
  theme = "colorblind",
  vsize = 11,
  asize = 5,
  mar = rep(6,4),
  layout = "circle")
```

temporal



contemporaneous



With the `graphicalVARsim()` function from the `graphicalVAR` package we can simulate model implied time series data based on the graphical VAR model we just created. This function requires us to specify the length of the time series we want to generate $nTime$, and our model implied β and κ matrix.

In the following example data with $t = 100$ is generated based on the graphical VAR model we created earlier:

```
var_data <- graphicalVARsim(nTime = 100, beta = true_model$beta, kappa = true_model$kappa)
```

Question 10

Now that you know how you can simulate data from a graphical VAR model, simulate various lengths of time series data $t = 50, 100, 500$ given our generated graphical VAR model. Estimate a graphical VAR model using the `graphicalVAR` function based on these simulated time series data. Plot the resulting networks and inspect the networks (Again, as we are visually comparing the networks don't forget to create the same layout for all networks we want to compare and maximum.). Are we able to estimate a similar network structure to the original network when we have time series data of length 50? What about when we have 500 data points?